Making Sense of Hybrid Access and MLO in IEEE 802.11ax/be

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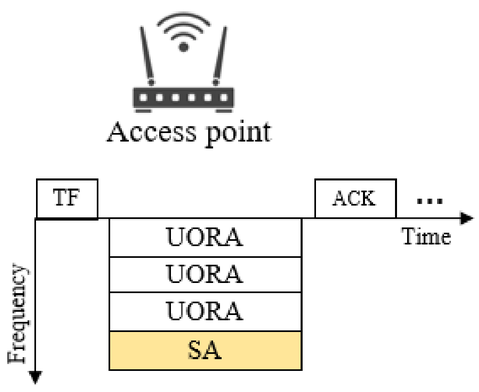
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Wi-Fi: Technological Enhancements

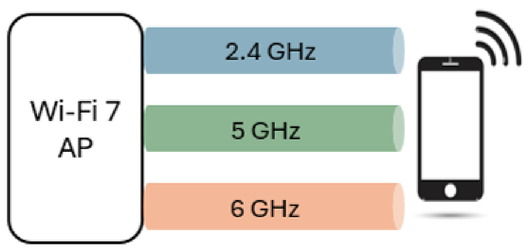
IEEE 802.11ax/be

- Physical layer: OFDMA and MU-MIMO
  - Enable concurrent transmissions
- Medium access control layer: hybrid access
  - Uplink OFDMA random access (UORA)
  - Scheduled access (SA)



IEEE 802.11be

- Multi-link operation (MLO)
  - Device can transmit and receive on multiple radio links

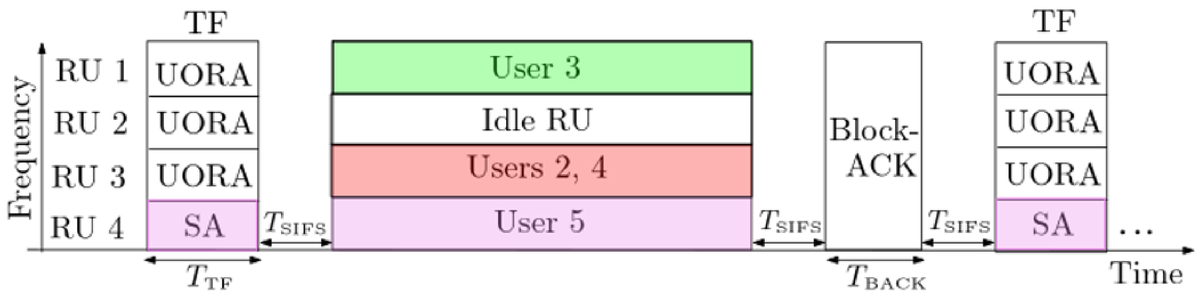


Goal: optimally use the key features together

Hybrid Access Protocol

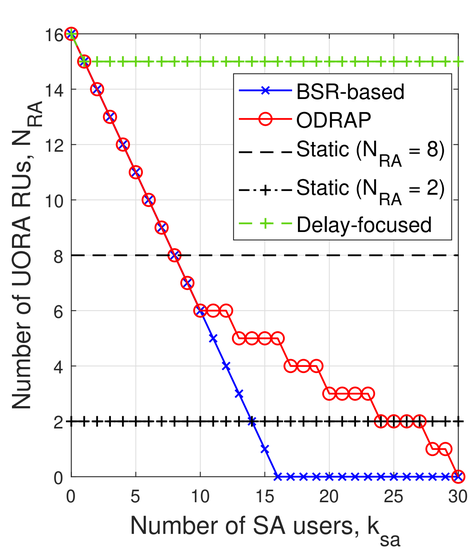
AP controls transmissions in hybrid access through trigger frames

- Scheduled access: AP schedules users in SA RUs
- Random access: users contend in UORA RUs to send BSRs



RU allocation trade-off

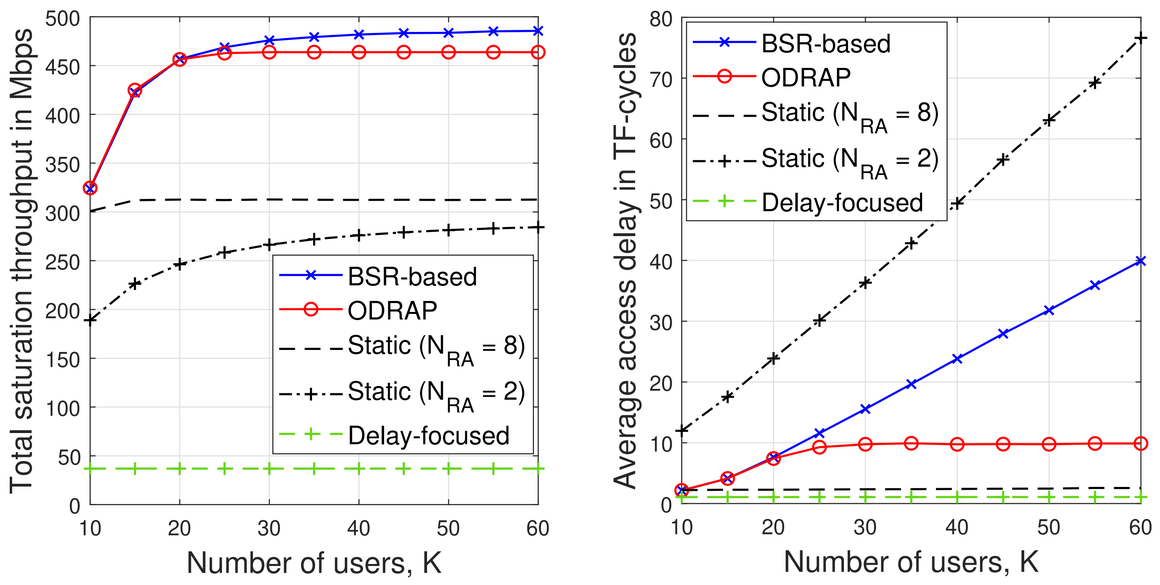
- More SA RUs: increases throughput
  - Increases access delay
- More UORA RUs: reduces delay
  - Reduces throughput



Optimal dynamic policy design

- RU allocation depends on SA users

Optimal Policy (ODRAP): Saturation Throughput and Access Delay



(160 MHz bandwidth, 16 RUs, 5 ms payload duration, and average packets in BSR = 10)

ODRAP optimally balances saturation throughput and access delay

Publications:

1. S. Arthi and N. B. Mehta, "Saturation throughput analysis of hybrid access MAC protocol in IEEE 802.11ax WLANs," in *Proc. GLOBECOM*, Dec. 2024, pp. 1–6.
2. S. Arthi, N. B. Mehta, and C. Singh, "Hybrid Access MAC Protocol in Wi-Fi: Analysis and Optimal Resource Allocation Policy Design," to appear in *IEEE TMC 2025.*

Multi-link Operation

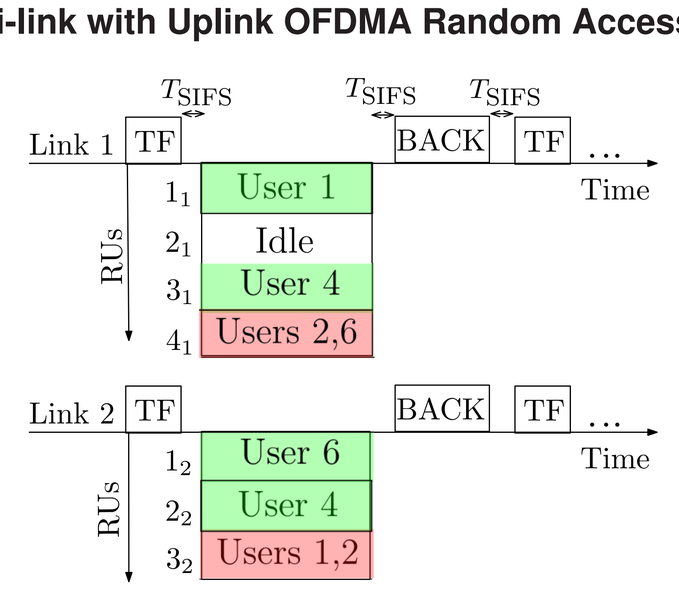
Multi-link operation: different modes

- Simultaneous MLO: simultaneous transmit-receive
  - Hardware is challenging due to power leakage between links
- Non-simultaneous MLO: transmit-transmit and receive-receive

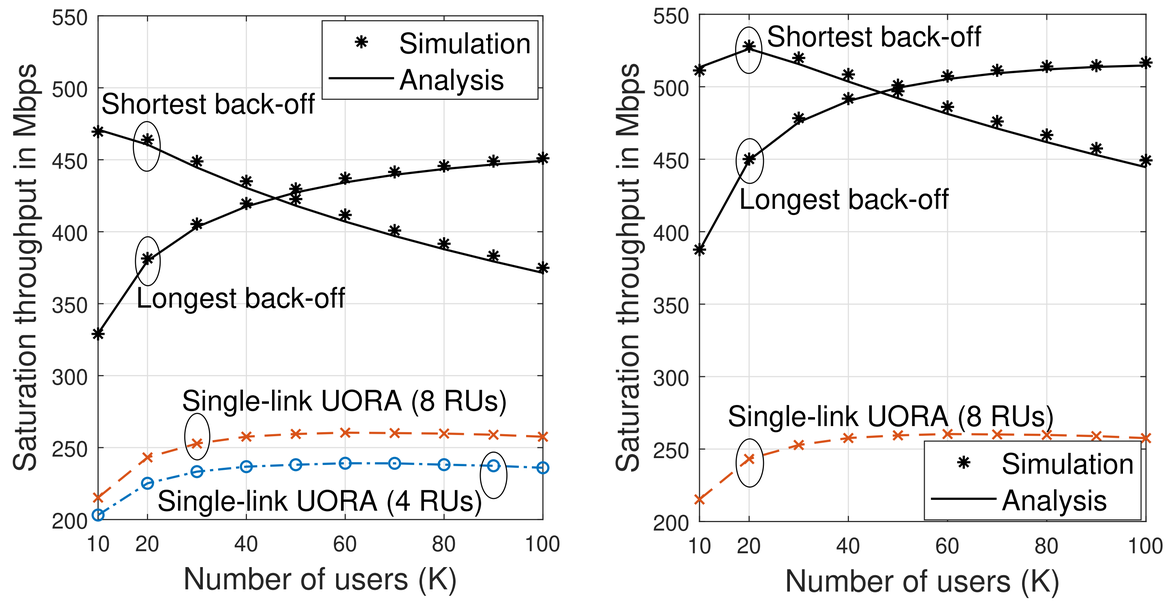
Non-simultaneous multi-link: two possible back-off rules

- Shortest back-off: device transmits if counter of any link is zero
- Longest back-off: device transmits if counters of both links are zero

Multi-link with Uplink OFDMA Random Access



Multi-link with UORA: Saturation Throughput



(8 RUs in link 1, 4 RUs in link 2)

(8 RUs link in 1, 8 RUs in link 2)

Multi-link UORA has a higher throughput than single-link UORA

Publications:

1. S. Arthi and N. B. Mehta, "Saturation Throughput Analysis of UORA-Enabled Multi-Link Operation in IEEE 802.11be Wi-Fi," submitted to *IEEE ICC 2026.*
2. S. Arthi and N. B. Mehta, "Combined Impact of PHY and MAC Layers on Throughput of Hybrid Access in Time-Varying Channels," submitted to *IEEE ICC 2026.*